

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2010

3423142

MATHEMATICS

Time : 2 hour

READ THESE INSTRUCTION CAREFULLY

1. You have been given this question booklet and separate answer sheet . The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER**YOUR NAME****NAME OF YOUR SCHOOL**

6. By drawing a dark line inside the correct numbered boxes mark your full Index number (i.e. School Code Number and the three figure Candidate's Number) in the grid near the top of answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the answer sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answer are lettered A,B,C and D. In case only **ONE** answer is correct choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example**In the question booklet:**

11. What is the value of $\frac{6(24-18)}{6} + 6 \times 4$?

6

- A. 30
- B. 25
- C. 10
- D. 28

1 [A] [B] [C] [D]

11 [A] [B] [C] [D]

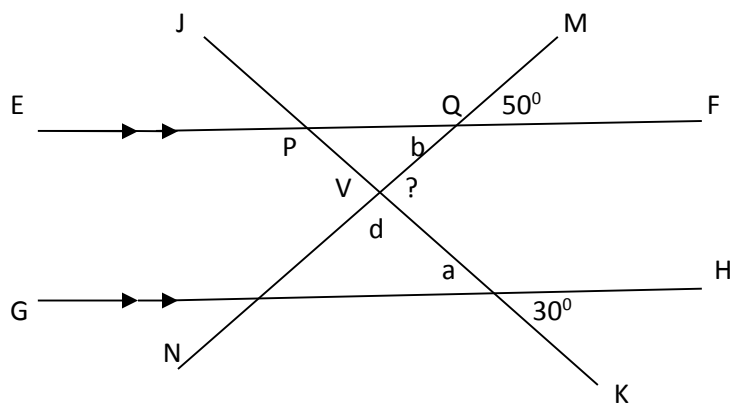
21 [A] [B] [C] [D] 31 [A] [B] [C] [D]

In the set, the box with the letter D printed in it marked.

12. Your dark line **MUST** be within the box.
13. For each question **ONLY ONE** box is to be marked in each set of the four boxes.

1. Which one of the following is 60400502 in words?
 - A. Six million four hundred thousand five hundred and two.
 - B. Sixty million four thousand five hundred and two
 - C. Sixty million fourty thousand five hundred and two
 - D. Sixty million four hundred thousand five hundred and two
2. What is the number 5826.3404 rounded off to 3 decimal places?
 - A. 5826.34
 - B. 5826.340
 - C. 5826.341
 - D. 5826.310
3. How many days are there between 15th July and 15th September?
 - A. 60
 - B. 61
 - C. 62
 - D. 63
4. What is the place value of digit 6 in the number 706053
 - A. Six hundreds
 - B. Hundreds
 - C. Six thousands
 - D. Thousands
5. What is the value of

$$\frac{3(4^2 + 2^2) - 5 \times 6 \div 2}{3 \times 5}$$
 - A. 59
 - B. 17
 - C. 11
 - D. 3
6. Teckla bought the following items from a shop.
3kg of sugar @ sh 68
250g of tea leaves for sh. 85
2 bars of soap @ sh. 38
1kg of cooking fat for sh. 109
2kg packet of rice for 149
 Teckla paid for the items using sh. 1,000note. How balance did he receive?
 - A. Sh. 220
 - B. Sh. 377
 - C. Sh. 551
 - D. Sh. 6
7. In the figure below, line EF are parallel. Line JK and MN are transversals which intersect at V.
 Angle MQF = 50° and angle HSK = 30°



What is the size of angle QVS?

- A. 150°
- B. 130°
- C. 100°
- D. 80°

8. Which one of the number below is the square of $2\frac{4}{5}$?

- A. $7\frac{21}{25}$
- B. $4\frac{16}{25}$
- C. $39\frac{1}{5}$
- D. $5\frac{3}{5}$

9. What is the next number in the pattern

10, 11, 15, 24, 40, 65, _____?

- A. 105
- B. 101
- C. 90
- D. 74

10. What is the simplest form of $5x + \frac{1}{4}(8x - 2y)$?

- A. $37x - 8y$
- B. $7x - \frac{1}{2}y$
- C. $28x - 2y$
- D. $7x - 2y$

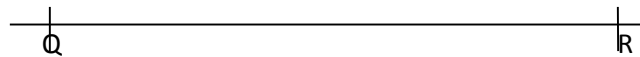
11. Below is a bus timetable from town J to town P.

TOWN	ARRIVAL TIME	DEPARTURE TIME
J	7:00 a.m	
K	9.30 a.m	10.00 a.m
L	11.15a.m	11.30a.m
M	12.15a.m	12.25p.m
N	1.10p.m	1.20p.m
P	1.50 p.m	2.00p.m

12. The perimeter of a rectangular plot of land is 280metres. The width of the plot is 60 metres. What is the length of the plot?

A. 70m
 B. 80.
 C. 110m
 D. 160m

13. Point S and line QR in the space below. Using a pair of compasses, drop a perpendicular from point S To meet line QR at T.



What is the length of line ST?

A. 2.8cm
 B. 3.5cm
 C. 4.5cm
 D. 5.5cm

14. What is the value of $\frac{2}{5} \div 1\frac{2}{3}$ of $\frac{3}{4}$?

A. $\frac{8}{9}$
 B. $\frac{1}{2}$
 C. $\frac{9}{50}$
 D. $\frac{8}{25}$

15. Thumbos paid sh. 10 200 for a cupboard after getting a discount of 15%. What was the marked price of the cupboard?

A. Sh. 1 530
 B. Sh. 8 670
 C. Sh. 11 730

- D. Sh. 12 000
16. Three bells are set to ring out at intervals of 4 minutes, 6 minutes and 9 minutes respectively. If they all ring together now, after how many minutes will they ring together next?
- 12
 - 18
 - 38
 - 216
17. A rectangular container is 2m long, 0.9m wide and 2.5m high. The container has water to a height of 1.5m. How much more water in litres is needed to fill the container?
- 1 800
 - 2 700
 - 4 500
 - 1 000
18. The fractions $\frac{3}{7}$, $\frac{2}{5}$, $\frac{5}{8}$, $\frac{1}{2}$ are to be arranged from the smallest to the largest. Which one of the following is the correct order?
- $\frac{1}{2}$, $\frac{2}{5}$, $\frac{3}{7}$, $\frac{5}{8}$
 - $\frac{2}{5}$, $\frac{3}{7}$, $\frac{1}{2}$, $\frac{5}{8}$
 - $\frac{3}{7}$, $\frac{2}{5}$, $\frac{1}{2}$, $\frac{5}{8}$
 - $\frac{5}{8}$, $\frac{1}{2}$, $\frac{3}{7}$, $\frac{2}{5}$
19. Which one of the following properties is TRUE for both a square and a rhombus?
- Diagonals are equal
 - All angles are equal
 - Opposite angles add up to two right angle
 - Diagonals bisect at a right angles.
20. The number of birds observed in a certain area during certain months of the year are as shown in the table below.

MONTH	APR	MAY	JUN	JULY	AUG
NUMBER OF BIRDS	96	104	80	118	94

- Which one of the following numbers is the highest mean of the birds recorded into two consecutive months?
- 106
 - 99
 - 111
 - 100
21. Construct a triangle XYZ in which $XY = 7.2\text{cm}$, $YZ = 5.8\text{cm}$ and $ZX = 6.2\text{cm}$. What is the size of angle XYZ?
- 125°
 - 75°
 - 55°

D. 50^0

22. What is the value of x in the equation

$$\frac{1}{2}(x + 1) + \frac{1}{3}(2x - 1) = 5?$$

A. $4 \frac{1}{7}$

B. $4 \frac{2}{7}$

C. $4 \frac{3}{7}$

D. $\frac{4}{7}$

23. A is in the form of pyramid with a square bas. Which one of the following statement is TRUE of the number of faces, edge and verticals the packet has?

A. 4 faces, 6 edges and 4 vertices

B. 2faces,1 edges and 1 vertex.

C. 5faces,9 edges and 6 vertices.

D. 5faces, 8 edges and 5 vertices.

24. A factory hired 9 people to complete a piece of work in 15 hours. How many more hours did it take them to complete the work if 3 people did not turn up?

A. 30 hours

B. $22 \frac{1}{2}$ hours

C. 5 hours

D. $7 \frac{1}{2}$ hours

25. Which one of the following sets of measurements will form a right angled triangle when drawn?

A. 9cm 16cm,25cm

B. 10cm,24cm,26cm

C. 5cm,12cm,17cm

D. 7cm,2.4cm,2.5cm

26. A pickup was loaded with four cartons of fat and 60 bales of flour. Each carton contained twenty four 250g packets of fat. The mass of each empty carton was 500g. Each bale contained twelve 2kg packets of flour. What is the total load, in tonnes?

A. 1466

B. 146.6

C. 14.66

D. 1.466

27. Kamau bought a piece of land for 2 million shillings. He sub- divided it into 25 plots of equal area. He then sold all the plots and made 20% profit. What was the selling price for each plot?

A. Sh. 400 000

B. Sh. 96 000

C. Sh. 80 000

D. Sh. 16 000

28. The table below shows how Kigen utilizes his piece of land.

Purpose	Number of hectares
Homestead Maize	$\frac{3}{4}$
Maize Cultivation	$1\frac{1}{4}$
Tea Cultivation	1
Grazing	$1\frac{1}{2}$

Which one of the bar graphs below correctly represents the information above.

29. Irimu deposited sh. 10 000 in a financial institution that offered simple interest at the rate of 5% per annum. Ndege deposited sh. 10 000 in a bank that offered a compound interest had Ndege's money earned than irimu's after 2 years?
- Sh. 25
 - Sh. 1 000
 - Sh. 1 025
 - Sh. 2 025
30. At a sports meeting the number of men was 200. The number of girls was three times that of men and 120 more than that of women. The number of boys was 30 more than that of girls. What was the total number of people at the meeting?
- 719
 - 1 850
 - 1 910
 - 2 150
31. The cash price of a radio was sh. 4 500. The hire purchase price of a radio was 60% more than the cash price. Muya bought a radio on hire purchase terms. He paid a deposit and 12 equal monthly instalments of sh. 540 each. How much did he pay as a deposit?
- Sh. 720
 - Sh. 6 480
 - Sh. 6 660
 - Sh. 7 200
32. A rectangle 25cm long and 12cm wide has the same area as a triangle whose height is 10cm. What is the length of the base of the triangle?
- 15cm

- B. 30cm
C. 60cm
D. 300cm
33. What is the value of $0.77 + 5.00$ of $(0.57 - 0.33) + 0.88 \times 0.4$?
- A. 2.322
B. 1.7368
C. 1.140
D. 0.90592
34. A salesman is paid a salary of sh. 5 000 per month. He is also paid a 2.5% commission on the sales above sh. 100 000. If the salesman sold goods worth sh. 500 000 in a certain month. What was his total earnings?
- A. Sh. 10 000
B. Sh. 12 500
C. Sh. 15 000
D. Sh. 17 500
35. On a map whose scale is 1 : 50 000, a piece of land is represented by a rectangle measuring 3cm by 12cm. What is the actual size of the land in hectares?
- A. 15
B. 150
C. 1 500
D. 15 000
36. Three schools Mwangaza, Kivuli and Nuru received a total donation of 165 textbooks. Kivuli got 8 books more than Mwangaza, while Nuru got half the total of what Mwangaza and Kivuli got. If the number of the books donated to Mwangaza is represented by letter m , which one of the following equations can be used to get the value of m ?
- A. $6m + 24 = 165$
B. $1\frac{1}{2}m + 12 = 165$
C. $3m + 12 = 165$
D. $3m - 12 = 165$
37. At the beginning of the year 2005, there were 800 pupils in a school of whom 55% were boys. At the end of the year the number of girls had increase by 20% and that of boys had decrease by 10%. What was the total number of pupils in the school at the end of the year?
- A. 828
B. 916
C. 828
D. 880
38. The height of isosceles triangle is 4cm. Each of the two equal sides measures 5cm. What is the area of the triangle?
- A. 6cm^2
B. 12cm^2
C. 15cm^2

D. 24cm^2

39. The table below shows the number of crates of soda Mutuma sold in one week. The number of crates sold on Friday was not recorded

DAYS OF WEEK	MON	TUE	WED	THUR	FRY	SAT	SUN
NUMBER OF CRETES	8	10	11	18	—	16	8

If the total number of crates of soda sold in the seven days was 84. What was the median sale?

- A. 13
B. 12
C. 11
D. 8
40. A teacher had a certain number of books. She gave $\frac{1}{3}$ of the books to John and $\frac{1}{4}$ to Lucy. She also gave $\frac{1}{10}$ of the remaining of the books to Patel. If the teacher was left with 18 books, How many books had she gave to Lucy?
- A. 48
B. 16
C. 12
D. 2
41. The perimeter of a rectangle is 24cm. The length of a rectangle is 2cm more than the width. What is the area of a rectangle?
- A. 15cm^2
B. 20cm^2
C. 35cm^2
D. 143cm^2
42. A cyclist took 15 minutes to travel from his home to town at a speed of 18km/h. He took 24 minutes to travel back from town to his home. What was the speed, in km/h, from town to his home?
- A. $1\frac{4}{5}$
B. $4\frac{1}{2}$
C. $11\frac{1}{4}$ 14 $\frac{8}{13}$
D. 4
43. Sera shared part of her land among her four children. Their share were 0.29, 0.26, 0.21 and 0.14 of the land. If the part that was shared was 36 hectares, how many hectares of the land remained?
- A. 3.6
B. 40
C. 0.1
D. 4
44. A farmer harvested 144 bags of maize in one season. In the second season the yield increased in the ratio 4 : 3. The farmer supplied all the bags of maize harvested in the second season equally to three millers. How many bags of maize did each miller get?

- A. 192
- B. 64
- C. 48
- D. 36

45. A rectangular water tank whose base is 1.5m by 0.5m is to be filled with water using 50 litres container. How many such containers will be required to fill the tank to a height of 1 metres?

- A. 15
- B. 1.5
- C. 150
- D. 1500

46. A watch loses 30 seconds every hour. If the watch was set right on Saturday at 11.30 p.m. What day and time did it show after 10 hours?

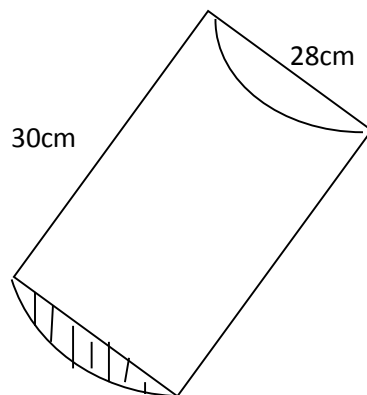
- A. Monday 9.25a.m
- B. Monday 9.30a.m
- C. Monday 9.35 a.m
- D. Monday 9.25 p.m

47. The graph below shows the journeys of two motorists Karimi and Nzomo.

How far from town X was Nzomo when Karimi stopped to rest?

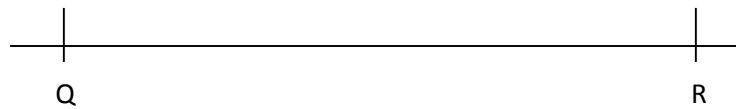
- A. 60km
- B. 185km
- C. 200km
- D. 250km

48. What is the surface area of the solid below in cm^2 ?



- A. 4096
- B. 2776
- C. 2468
- D. 1936

49. On the QR given below, construct a triangle PQR such that $PQ = PR = 7\text{cm}$. Construct a bisector of angle PQR to meet line PR at X.



What is the size of angle QXR?

- A. 78°
- B. 44°
- C. 68°
- D. 102°

50. The figure shows a pattern of shapes.



Which one of the shapes below should be drawn in the black box to continue with the pattern?

